# Jacob Dineen

AI, Reasoning & Cognition (ARC) Lab — Arizona State University

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## RESEARCH INTERESTS

I work on reasoning and alignment in large language models, with related interests in multi-agent reinforcement learning, controllability, and explainable AI (XAI). I have also held data science, ML engineering, and applied research roles across financial services and AI startups.

## EDUCATION

**Arizona State University — Ph.D. Artificial Intelligence** *GPA 4.00/4.00 · Advisor: Ben Zhou* 2022–2027

*Committee: Muhao Chen, Chitta Baral, Vivek Gupta*

**University of Virginia — M.Sc. Computer Science** *GPA 3.96/4.00 · Advisor: Madhav Marathe* 2019–2021

**Syracuse University — M.S. Data Science** *GPA 4.00/4.00* 2017–2018

**Grand Canyon University — B.S. Finance & Economics** *GPA 3.65/4.00* 2012–2015

## RESEARCH EXPERIENCE

**Graduate Research Assistant, ARC Lab (Arizona State University)** 2024–Present

- Research on reasoning, alignment, controllability, and multi-agent RL in LLMs.

**Graduate Research Assistant, SEFCOM (Arizona State University)** 2022–2024

- AI and cybersecurity research.

**Applied Research, Capital One — Center for Machine Learning (C4ML)** 2020–2021

- Built reinforcement learning and agent-based simulations for organizational dynamics.

**Research Assistant, Biocomplexity Institute (University of Virginia)** 2019–2020

- Worked on graph dynamical systems, cooperative game theory, and behavioral modeling.

## PROFESSIONAL EXPERIENCE

**AI/ML Engineer, Alumbra AI** Part-Time 2026

**Research, Pareto AI** Part-Time 2025–Present

**Machine Learning Engineer, Spring Oaks Capital** Full-Time 2022–2025

**Data Scientist, Capital One** Full-Time 2020–2022

**Analyst & Business Intelligence, Real World Marketing** Full-Time 2016–2019

**Data Scientist, Buffalo Check LLC** Part-Time 2015–2019

**Optimization Analyst, Voltari** Part-Time 2012–2015

## PUBLICATIONS

Google Scholar Profile

[+] Corresponding Author / Mentor

### Peer-Reviewed Conference Proceedings (C)

C1. Ye, Xiao, Li, Zhaonan, **Dineen, Jacob**, Xu, Zhikun, Lu, Shijie, Shen, Ming, Shrivastava, Shaswat, Ahuja, Avneet, & Zhou, Ben (2026). *Learning Verifiable Reasoning Programs through Cohort Consistency*. In *Proceedings of the 2026 Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the International Joint Conference on Natural Language Processing* (*AACL-IJCNLP 2026*).

C2. Shen, Ming, Xu, Zhikun, Ye, Xiao, **Dineen, Jacob**, & Zhou, Ben (2026). *BOW: Training Language Models to Reason Over Plausible Next Words*. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*.

C3. Xu, Zhikun, Feng, Yu, **Dineen, Jacob**, Shi, Taiwei, Zhao, Jieyu, & Zhou, Ben (2026). *Skill Reuse as Compression in Agentic RL*. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*.

C4. **Dineen, Jacob**, RRV, Aswin, Xu, Zhikun, & Zhou, Ben (2026). *Vocabulary Dropout for Curriculum Diversity in LLM Co-Evolution*. In *Proceedings of the Conference on Language Modeling* (*COLM 2026*).

C5. **Dineen, Jacob**[+], RRV, Aswin, Liu, Qin, Xu, Zhikun, Ye, Xiao, Shen, Ming, Li, Zhaonan, Lu, Shijie, Baral, Chitta, Chen, Muhao, & Zhou, Ben (2025). *QA-LIGN: Aligning LLMs through Constitutionally Decomposed QA*. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 20619–20642.

C6. RRV, Aswin, **Dineen, Jacob**, Handa, Divij, Uddin, Md Nayem, Parmar, Mihir, Baral, Chitta, & Zhou, Ben (2025). *ThinkTuning: Instilling Cognitive Reflections without Distillation*. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 31236–31250.

C7. Xu, Zhikun, Shen, Ming, **Dineen, Jacob**, Li, Zhaonan, Ye, Xiao, Lu, Shijie, RRV, Aswin, Baral, Chitta, & Zhou, Ben (2025). *ToW: Thoughts of Words Improve Reasoning in Large Language Models*. In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* (*Volume 1: Long Papers*), pp. 3057–3075.

C8. **Dineen, Jacob**[+], Kridel, Donald J., Dolk, Daniel R., & Castillo, David G. (2023). *Unified Explanations in Machine Learning Models: A Perturbation Approach*. In *Proceedings of the 56th Hawaii International Conference on System Sciences* (*HICSS-56*), pp. 795–804.

C9. **Dineen, Jacob**[+], Haque, A. S. M. Ahsan-Ul, & Bielskas, Matthew (2021a). *Formal Methods for an Iterated Volunteer's Dilemma*. In *Proceedings of the 14th International Conference on Social, Cultural, and Behavioral Modeling* (*SBP-BRiMS 2021*), pp. 81–90.

C10. **Dineen, Jacob**[+], Haque, A. S. M. Ahsan-Ul, & Bielskas, Matthew (2021b). *Reinforcement Learning for Data Poisoning on Graph Neural Networks*. In *Proceedings of the 14th International Conference on Social, Cultural, and Behavioral Modeling* (*SBP-BRiMS 2021*), pp. 141–150.

C11. Dolk, Daniel R., Kridel, Donald J., **Dineen, Jacob**[+], & Castillo, David G. (2020). *Model Interpretation and Explainability towards Creating Transparency in Prediction Models*. In *Proceedings of the 53rd Hawaii International Conference on System Sciences* (*HICSS-53*).

## Peer-Reviewed Workshop Papers (WS)

WS1. Srinivasan, Adarsh, **Dineen, Jacob**$^{+}$, Sarfraz, Muhammad Uzair, Afzal, Muhammad Umar, Riaz, Irbaz, Zhou, Ben (2026). *RECAP: Transparent Inference-Time Emotion Alignment for Medical Dialogue Systems*. In *Proceedings of the 8th Workshop on Clinical Natural Language Processing (Clinical NLP) @ LREC 2026*, pp. 350–368. Oral

WS2. Li, Zhaonan, Chickering, Kyle R., Li, Bangzheng, **Dineen, Jacob**, Ye, Xiao, Xu, Zhikun, Lu, Shijie, Huang, Yuxi, Shen, Ming, Nguyen, Bach, Pavuluri, Jaya Adithya, Nguyen, Mau Son, Chavan, Sanika, Le, Ngoc Minh Thu, Chen, Muhao, Zhou, Ben (2026). *VisAnalog: A Diagnostic Suite for Visual Concept Transfer on Natural Images*. CVPR Workshop on Visual Concepts (VisCon), 2026. Oral

## Working Papers / Under Review (W)

W1. **Dineen, Jacob**, Ren, Silei, Chen, Muhao, Roth, Dan, & Zhou, Ben (2026). *Despite Instructions: Frontier Agents Improvise Covert Channels at Test Time*. Under Review, ICLR 2027

W2. Li, Zhaonan, Lu, Shijie, Wang, Fei, **Dineen, Jacob**, Ye, Xiao, Xu, Zhikun, Liu, Siyi, Cho, Young Min, Li, Bangzheng, Chang, Daniel, Nguyen, Kenny, Yang, Qizheng, Chen, Muhao, Zhou, Ben (2025). *Unbiased Visual Reasoning with Controlled Visual Inputs*. arXiv Preprint

W3. Liu, Qin, **Dineen, Jacob**, Huang, Yuxi, Zhang, Sheng, Poon, Hoifung, Xiao, Chaowei, Zhou, Ben, Chen, Muhao (2025). *ArenaBencher: Automatic Benchmark Evolution via Multi-Model Competitive Evaluation*. Under Review, ICLR 2027

W4. RRV, Aswin, **Dineen, Jacob**, Handa, Divij, Parmar, Mihir, Zhou, Ben, Baral, Chitta, Mishra, Swaroop (2026). *Mid-training with Self-Generated Data Improves Reinforcement Learning in Language Models*. arXiv Preprint

W5. Ye, Xiao, **Dineen, Jacob**, Zhu, Evan, Lu, Shijie, Song, Kevin, Zhou, Ben (2026). *HINDCAST: Replaying Prediction Markets to Evaluate LLM Forecasters*. arXiv Preprint

W6. Ye, Xiao, **Dineen, Jacob**, Li, Zhaonan, Xu, Zhikun, Chen, Weiyu, Lu, Shijie, Huang, Yuxi, Shen, Ming, Tran, Phu, Yum, Ji-Eun Irene, Khan, Muhammad Ali, Afzal, Muhammad Umar, Riaz, Irbaz Bin, Zhou, Ben (2025). *Evaluating Medical LLMs by Levels of Autonomy: A Survey Moving from Benchmarks to Applications*. arXiv Preprint

## SKILLS

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|---|---|
| **LLM Methods** | pre/mid/post-training, RL (GRPO, RLHF), reward modeling, self-play, agentic tool use, RAG, synthetic data generation, evaluation and benchmarking |
| **ML Libraries** | PyTorch, TensorFlow, JAX, Polars, Dask, HuggingFace, TRL, vLLM, VeRL, PySpark, NetworkX, DGL, Torch Geometric, BoTorch, SnowparkML, Anthropic/OpenAI SDKs |
| **Languages** | Python, Rust, C, C++, Java, JavaScript, R, Bash, PRISM, x86-64 |
| **Databases** | MySQL, PostgreSQL, MongoDB, Snowflake, Redis, SQLite, Redshift |
| **Tools** | Git, GitHub Actions, Docker, Kubernetes, Helm, Airflow, AWS (ECS, ECR, S3, EKS, CodeBuild), SageMaker, Databricks, FastAPI, Next.js, pytest, OR-Tools, Sigma, Streamlit |

## SERVICE

**Conference Reviewer:** HICSS, SBP-BRiMS, NAACL, EMNLP, EACL, COLM

**Teaching / Tutoring:**

- CGCC Calculus & Linear Algebra Tutor (2019)
- Teaching Assistant — ASU CSE 365 / pwn.college (Security)

## MISCELLANEOUS

**Expert AI Trainer, Pareto AI** (2024–Present)

**Independent Contractor, Mercor Intelligence** (2025–Present)

**Cybersecurity:** pwn.college green belt (binary exploitation), user: `jdin`

## GRADUATE COURSEWORK

**Arizona State University — PhD Computer Science**:
Knowledge Representation; Computer Systems Security; Software Security; Planning and Learning Methods in AI; Algorithms; Research/Dissertation Hours.

**University of Virginia — MSc Computer Science**:
Algorithms; Machine Learning; Computer Vision; Formal Methods; Reinforcement Learning; Graph Mining; Learning Theory (Game Theory); Cloud Computing; Research Hours.

**Syracuse University — MS Data Science**:
Data Analysis and Decision Making; Business Analytics; Financial Analytics; Marketing Analytics; Advanced Information Systems; Data Science; Data Warehousing; Text Mining; Scripting for Data Analysis; Information Policy.